United Arab Emirates
Artificial Intelligence Strategy

Stakeholder workshop

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DUBAI FUTURE FOUNDATION

Workshop Objectives

Help shape the UAE's AI strategy

A – Confirm the UAE's starting point and aspiration for AI in five years time

B – Agree on the most important opportunities to realize the UAE's AI vision

C – Shape initiatives in the strategy

Agenda – AI Strategy workshop

5 February 2018, 2:00-4:30 PM, Office of the Future

Time	Topic	Speaker
1:45pm – 2:00pm	Registration, tea and coffee	
2:00pm – 2:10pm	**Workshop start** Welcome, purpose and introductions	• Kate Pounder
2:20pm – 2:25pm	**Overview of AI strategy**	• Jessica Bland
2:25pm – 3:25pm	**Setting aspiration for AI in the UAE and prioritizing opportunities**	• Kate Pounder
3:25pm – 4:25pm	**Shaping AI initiatives** *Station 1:* *Innovate Government services* *Station 2:* *Boost Industry* *Station 3:* *Enable workforce* *Station 4:* *Lift R&D capability* *Station 5:* *Adopt effective regulation and governance*	• Corin Moffatt
4:25pm – 4:30pm	**Workshop conclusion** Wrap up and next steps	• Jessica Bland and Kate Pounder

AI strategy overview

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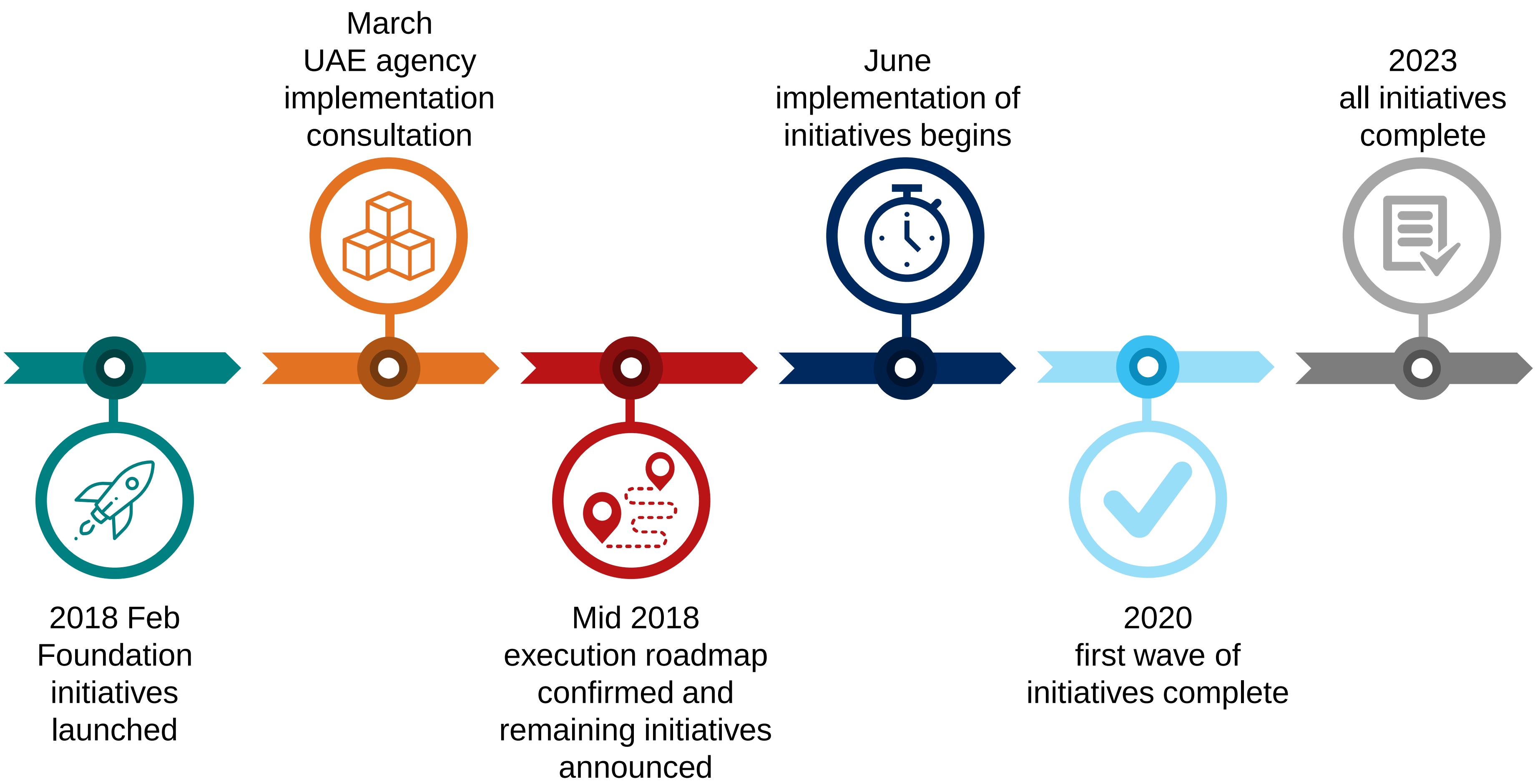
The UAE is developing a five-year strategy to make it an AI world-leader by 2031



The UAE is proposing to focus on five objectives to achieve it's vision of becoming an AI world leader

Vision

To be an AI world-leader by 2031

Mission

Transforming UAE into a world leader in AI by investing in the people and industries that are key to our success

Objectives

1. Attract and train people for future jobs enabled by AI
2. Bring world-leading research capability in key research sectors to UAE
3. Support industry to develop and deploy AI expertise to sustain global position
4. Transform government services to improve efficiency and customer experience
5. Build an 'AI destination' reputation that balances innovation, enabling regulation and ethics through good governance

The AI strategy will ultimately provide 100 initiatives across 23 areas of opportunity and five strategic pillars to achieve UAE's vision

Vision
To be an AI world leader by 2031

Mission
Transforming UAE into a world leader in AI by investing in people and industries that are key to our success

Transform government services to improve efficiency and customer experience	Support industry to develop and deploy AI expertise to sustain global position	Attract and train people for AI-fueled jobs for the future	Develop world-leading research capability in key research sectors	Build an 'AI destination' reputation that balances enabling regulation and ethics through good governance

Strategic Pillars

Innovate government	**Boost industry**	**Enable workforce**	**Lift R&D capability**	**Adopt effective governance**

Opportunities

Use analytics and optimization to improve policy and operations	Support trials of innovative applications of AI	Identify and manage future workforce needs	Coordinate research strategy	Regularly consult with business and society to monitor emerging trends
Deploy autonomous systems and remote sensing	Use Gov procurement and incentives to accelerate AI adoption and development	Undertake workforce based training	Build research capability and attract talent	Design regulation that enables AI innovation
Improve citizen-facing service delivery	Identify and expand AI export opportunities	Develop skilled talent through tertiary education and vocational training	Create common R&D infrastructure	Create ethical standards for the application of AI
Roll-out enabling services and infrastructure to support innovation	Attract foreign investment in the UAE AI sector	Attract skilled talent via immigration	Provide direct and indirect funding for R&D	Engage in international cooperation to help set AI standards
		Ensure schools provide future-ready education	Strengthen commercialisation of AI technology	Ensure appropriate cybersecurity capabilities

Initiatives

Flagship and other phase 1 initiatives

Remaining initiatives

UAE vision, mission and objectives

What the strategy will achieve

5 pillars

Areas Government can take action in

23 opportunities

The potential AI opportunities within each pillar

~100 initiatives

The specific actions Government will take

The strategy will develop a new generation of talent to advance the use of core AI technologies in the UAE

Data analysis

Acquisition of data and transformation into actionable insight or foresight

Examples

- Computer vision, hearing and language processing
- Anomaly detection, collaborative filtering, motion and path planning and pattern/motif recognition
- Clustering, deep and reinforcement learning, dimensionality reduction, networks/graphical models, pattern similarity, probabilistic modelling, regression and search algorithms

Automated systems

Manage complex physical and digital systems, including responding to disruptions and unfamiliar inputs.

Examples

- Fixed and mobile land, underwater and aerial robots
- Self-driving transportation systems
- Wireless communication and information networks

Digital interface

Simulate behavior to interact meaningfully with humans and exchange information.

Examples

- Virtual and augmented reality
- Digital twins and virtual assistants
- Wearable and assistive devices
- Deep learning and neural nets
- Cloud computing and edge device hardware

SOURCE: AlphaBeta analysis, expert panel input

The strategy will facilitate the opportunity to apply ground-breaking AI innovations at scale

		Examples of application		
	Key potential outcomes	Data analysis	Automated systems	Digital interface
Transport and Logistics	▪ Reduce congestion and travel times ▪ Improve urban and network planning ▪ Improved safety	▪ Optimise logistics routes	▪ Self-driving cars	▪ Navigation assistance
Health	▪ Preventing leading causes of ill health ▪ Reduce cost of health care ▪ Reduce error in diagnosis and treatment	▪ Pathology and diagnosis	▪ Surgical assistants	▪ Remote medicine and telehealth
Space	▪ Optimise design and manufacture of advanced technology including space craft ▪ Analyse vast amounts of visual and other data	▪ Satellite photo analysis ▪ Test experiments	▪ Spacecraft and exploration robotics	▪ Software to control spacecraft
Energy	▪ Improve efficiency of product design ▪ Decrease energy waste and leakage ▪ Maintain infrastructure at lower cost	▪ Optimise asset operation	▪ Asset inspection	▪ Augmented reality for safety
Education	▪ Expand access to high-quality education ▪ Reduce cost of education services ▪ Improved student engagement	▪ Identify variables of student performance	▪ Virtual reality systems for testing/ trials	▪ Machine tutors and e-teachers
Safety and security	▪ Improve crime prevention ▪ Improved emergency response times ▪ Reduce cost and risk of surveillance	▪ Predictive policing	▪ Unmanned aerial vehicles	▪ Digital emergency response dispatch
Environment	▪ Improve air and coastal water quality monitoring ▪ Locate water sources more efficiently ▪ Lower cost of maintaining and repairing physical infrastructure	▪ Review surveillance to enforce laws	▪ Asset inspection	▪ Digital platform to report env. concerns

SOURCE: AlphaBeta review of AI applications, expert panel input

It is proposed to launch AI initiatives in phases that progressively cement the UAE's position as a world-leader in AI

Phase 1

Jan – Jun 18

Lay foundations

Announce key initiatives in areas:

- That are pre-requisites for the whole strategy's success
- Have the agreement of key stakeholders
- Warrant exploration, where identified as a priority but require more work to agree them

Phase 2

Jul-Dec 18

Accelerate advantage

Identify and commence roll-out of remaining 100 initiatives in areas that require stakeholder engagement

Phase 3

2019 – 2023

Progress to world leader

100 initiatives embedded and delivering results

During phase 2, UAE will work with government, industry and experts to commence implementation of the remaining 100 AI initiatives

Identify

Identify additional initiatives that will help UAE achieve its vision

Consult

Consult Government agencies, industry and experts to confirm support, feasibility and impact

Announce

Announce the remaining initiatives

Implement

Support responsible Government agency to design the implementation of initiatives and ensure effective governance is in place

Setting the aspiration and prioritizing opportunities

Exercise 1: Setting aspiration for AI in the UAE

Objective: Confirm the UAE's aspiration for AI in 3 – 5 years

Approach

- Mark with a green sticky dot where the UAE is currently on the opportunity scale
- Mark with a blue sticky dot where you think the UAE should aspire to be on the scale in 3-5 years
- Join in the group discussion

Workshop questions

- Where does the UAE aspire to be world-leading in 3-5 years?
- What were the reasons for choosing these areas?
- What will it take to become world-leading?
- Which areas should not be prioritized?
- What is the reason for not prioritizing them?
- What are examples of ground-breaking projects the UAE could undertake?

Exercise 1: Setting the UAE's aspiration for AI

Across each opportunity, assess UAE's current capacity and its potential to improve over the next 3 – 5 years	Beginning	Progressed	Advanced
	Limited AI skills and use (0-2) projects/pilots	*Experimenting with adopting AI (a few projects/more complex pilots), some workforce capacity*	*Using leading edge technologies and ground-breaking approaches to AI at scale*
Use analytics and optimization to improve policy and operations			
Deploy autonomous systems and remote sensing			
Improve citizen-facing service delivery			
Roll-out enabling services and infrastructure to support innovation			
Support trials of innovative applications of AI analytics and optimization			
Support industries to deploy autonomous systems and remote sensing			
Identify and expand AI export opportunities			
Attract foreign investment in the AI sector			
Identify and manage future workforce needs			
Undertake workforce based training			
Develop domestic talent through tertiary education and vocational training			
Attract skilled talent via immigration			
Ensure schools provide future-ready education			
Coordinate research strategy			
Build research capability and attract talent			
Create common R&D infrastructure			
Provide direct and indirect funding for R&D			
Strengthen commercialization of AI technology			
Regularly consult with business and society to monitor emerging AI trends			
Design regulation that enables AI innovation			
Create ethical standards for the application of AI			
Engage in international cooperation to help set AI standards			
Ensure appropriate cybersecurity capabilities			

Scale of opportunity: Innovate Government Services

Opportunity	Beginning	Progressed	Advanced
Use analytics and optimization to improve policy and operation	Limited or no use of advanced AI software/ mechanisms for data analysis in government	Several initiatives introduced, a few are advanced or well-trialled	Sophisticated analytical systems and optimization initiatives introduced and used widely
Deploy autonomous systems and remote sensing	Limited or no use of autonomous systems and remote sensing	Trialling/operating sensors in some key service areas e.g. transport	Extensive use of sensors and systems across sectors, integrated in operation and planning
Improve citizen-facing service delivery	Limited or no automation of transactional services and digital service delivery	Some key government services automated e.g. payments, registration	End to end digitisation and automation of services
Roll-out enabling services and infrastructure to support innovation	Limited or no publicly available datasets and open-source platforms	Some ad-hoc sharing of data across agencies and with public to problem solve	Sophisticated data sharing and open software across Government and with public

Scale of opportunity: Boost Industry

Opportunity	Beginning	Progressed	Advanced
Support trials of innovative applications of AI analytics and optimization	Minimal or slow application of AI systems to improve productivity	Ad hoc application of AI systems in some critical sectors	Comprehensive support to transition industries toward AI systems
Support industries to deploy autonomous systems and remote sensing	Limited or no use of autonomous systems in local industries	Adoption of select autonomous systems in some major industries	Thorough adoption of autonomous systems by several industries
Identify and expand AI export opportunities	Limited or no focus on identifying and developing AI export markets	Some initiatives to find and connect to export markets in AI, but limited to only some geographies and sectors	Support across many sectors to identify and develop global export markets
Attract foreign investment in the UAE AI sector	Limited or no efforts to increase foreign investment into the AI sector	Some obstacles to foreign investment identified and addressed, with a focus on the AI sector	Proactive and coordinated national strategy to attract foreign investment in AI sector

Scale of opportunity: Enable workforce

Opportunity	Beginning	Progressed	Advanced
Identify and manage future workforce needs	Limited or no understanding of economy-wide shifts in workforce needs	Some partial understanding of workforce needs in key sectors or geographies	Thorough understanding of evolving workforce needs on different occupations across the economy
Undertake workforce based training	Limited or no programme to ensure sufficient workforce-based training for AI is provided by employers and undertaken by employees	Some national incentives to promote workforce-based training	Clear understanding of current workforce training gaps and coherent strategy to address them
Develop domestic talent through tertiary education and vocational training	Limited or no efforts to change vocational and tertiary education systems to adopt to AI	Some major courses responding to AI but not widespread	Coherent strategy to identify education/ training needs and implement across all tertiary and vocational institutions
Attract skilled talent via immigration	Successful at attracting some AI talent but limited in number and type of qualification	Qualified AI talent attracted in some key geographies and industries	Globally competitive destination for senior AI talent
Ensure schools provide future-ready education	Limited or no modification to school curriculum to respond to disruptions from AI	Some advanced computer literacy and technical skills introduced in schooling program	Comprehensive realignment of schooling to teach technical and complementary skills, as well as increase exposure to AI

Scale of opportunity: Lift research and development capability

Opportunity	Beginning	Progressed	Advanced
Coordinate research strategy	Limited or no understanding of research priorities for AI systems and technology	Informal or partial research strategy for AI systems and technology	Coordinated, national research strategy has been developed for AI systems and technology
Build research capability and attract talent	Minimal effort to focus on AI in developing or attracting research talent	Some investments to build research capability and attract talent in AI	Focussed, target-led approach to build research capability and attract talent
Create common R&D infrastructure	Limited or no efforts to establish common infrastructure to facilitate AI R&D	R&D infrastructure in early stages of development or planning	Common R&D infrastructure developed and deployed to facilitate AI research
Provide direct and indirect funding for R&D	Existing R&D funding initiatives not tailored to encourage AI systems and technology	Some programs exist to fund R&D into AI systems and technology	Long-term and target-led efforts to provide funding for R&D into AI systems and technology
Strengthen commercialization of AI technology	Limited or no ecosystem to facilitate commercialisation of AI technology	Basic support provided in key geographies or sectors for commercialisation of AI technology	Comprehensive national support for commercialisation of AI technology

Scale of opportunity: Adopt effective governance and regulation

Opportunity	Beginning	Progressed	Advanced
Regularly consult with business and society to monitor emerging AI trends	Limited or no consultation with business and society	Ad hoc consultation with some stakeholders in business and society	Comprehensive and regular consultation with key business and society stakeholders
Design regulation that enables AI innovation	Regulation not considered or modified for AI technology	Some updated regulation but with limited consideration of impact on AI innovation	Coherent approach to AI regulation which boosts innovation
Create ethical standards for the application of AI	Informal or no new ethical standards articulated to govern AI systems	Reactive or partial ethical standards designed to govern AI systems	Proactive and considered articulation of ethical standards to govern AI systems
Engage in international cooperation to help set AI standards	Limited international engagement to help set AI standards	Engagement in some international efforts to help set AI standards	Leading international engagement to drive cooperation to set AI standards
Ensure appropriate cybersecurity capabilities	Limited or no effort to ensure cybersecurity capabilities	Cybersecurity protections developed for certain key systems	Comprehensive and forward-looking national approach to developing cybersecurity capabilities

Exercise 2: What matters most to our vision?

Objective: Identify the most important opportunities for realising the UAE’s vision for AI

Approach

- Mark with a red dot the opportunities that are most important to realizing the vision and objectives for AI in the UAE
- Write on a post-it note any opportunities that are missing from the list
- Join in the group discussion

Workshop questions

- Which opportunities are most important to our vision and objectives?
- What was your reasoning in choosing these opportunities?
- What opportunities are of lesser importance? Why?
- What opportunities were missing and why do they matter?

Exercise 2: What matters most to our vision of becoming an AI world leader?

Strategy objectives	Areas of action	Opportunity	Priority
Transform government services to improve efficiency and customer experience	Innovate government services	Use analytics and optimization to improve policy and operations	
		Deploy autonomous systems and remote sensing	
		Improve citizen-facing service delivery	
		Roll-out enabling services and infrastructure to support innovation	
Support industry to develop and deploy AI expertise to sustain global position	Boost industry performance	Support trials of innovative applications of AI analytics and optimization	
		Support industries to deploy autonomous systems and remote sensing	
		Identify and expand AI export opportunities	
		Attract foreign investment in the AI sector	
Attract and train people for AI-fueled jobs for the future	Ensure workforce readiness	Identify and manage future workforce needs	
		Undertake workforce based training	
		Develop domestic talent through tertiary education and vocational training	
		Attract skilled talent via immigration	
		Ensure schools provide future-ready education	
Develop world-leading research capability in key research sectors	Lift R&D capability	Coordinate research strategy	
		Build research capability and attract talent	
		Create common R&D infrastructure	
		Provide direct and indirect funding for R&D	
		Strengthen commercialization of AI technology	
Build an 'AI destination' reputation that balances enabling regulation and ethics through good governance	Adopt effective governance and regulation	Regularly consult with business and society to monitor emerging AI trends	
		Design regulation that enables AI innovation	
		Create ethical standards for the application of AI	
		Engage in international cooperation to help set AI standards	
		Ensure appropriate cybersecurity capabilities	

Shaping initiatives

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Exercise 3: Shaping AI Initiatives

Objective: Identify the most important initiatives for realising the UAE's vision for AI

Approach

- Form five groups and move to the station that corresponds to your group
- In your group, answer the workshop questions
- After 20 mins, rotate to the next pillar and spend 5 mins building on the previous group's work
- Repeat this step for the remaining 3 pillars

Workshop questions

- Review the potential initiatives in your pillar:
  - Are these the appropriate flagship initiatives to announce for this pillar?
  - Would you amend the initiatives in any way?
  - Are there any initiatives you suggest adding / removing?
  - Are any initiatives already underway?
  - As individuals, use a red dot to indicate the top 3 most important initiatives in this pillar

Group 1	Group 2	Group 3	Group 4	Group 5
Innovate government	**Boost industry**	**Enable workforce**	**Lift R&D capability**	**Adopt effective governance**

Potential flagship initiatives

National AI challenge

The UAE government will establish a national challenge process, inviting Government agencies and AI experts to identify opportunities to use AI to solve UAE's most important challenges and providing prize incentives for the best ideas e.g. preventing leading causes of poor health, improving traffic flow

Possible initiatives to realise four key opportunities to innovate government services

A Innovate government services

1. Use analytics and optimization to improve policy and operations

- **Smart health:** e.g. better diagnose illnesses and monitor public health priorities and resource allocation
- **Trade & economics:** e.g. se AI systems to analyse broad economic data, identify trade opportunities, enhance forecasting etc.
- **Improved justice and policing:** e.g. AI analysis of crime statistics to enable appropriate allocation of resources and "preventative policing"
- **Smart emergency management:** e.g. use health data and crowd-sourced input to better direct emergency responses
- **Smart welfare and public payments:** e.g. ensure integrity of payment applications and regular payments through AI-enabled monitoring processes

2. Deploy autonomous systems and remote sensing

- **Traffic management:** e.g. install smart signals and sensors to improve management of road and pedestrian traffic
- **Automated public transport:** e.g. use autonomous vehicles and AI-software to provide driving assistance
- **Infrastructure maintenance:** e.g. use sensors and crowd-sourced input to identify priorities for infrastructure repair and maintenance
- **Climate monitoring:** e.g. distributed network of sensors to monitor the impact of waste disposal and other activity on ecological systems
- **Robotics for basic services:** e.g. establish an automated waste sorting system to enable more efficient waste management

3. Improve citizen-facing service delivery

- **Automate online services:** e.g. provide real-time responses for services provided through online systems by processing with AI-enabled systems
- **Digitise and automate transactions:** e.g. Partially automate building inspections through combining web-based image and data analysis with site visits
- **Efficient identity verification:** e.g. enable ID verification for credit checks or passport applications through automated analysis of broad data such as Photo ID, bank statements etc.

4. Roll out enabling services and infrastructure to support innovation

- **Data exchange:** e.g. establish a secure data exchange between Government agencies to promote select data sharing
- **Make data available:** e.g. publish data from traffic sensor network to select other agencies and public entities
- **Improved storage and networking:** e.g. transition Government information to "data oceans" enabled by secure shared servers
- **Data consolidation:** e.g. unify registries of public information to create comprehensive citizen services data
- **Training and development:** e.g. train Government employees in use and deployment of AI systems, and development of new solutions

For future consideration *Possible February announcement*

Potential flagship initiatives

Global Value Chains

The UAE government will work with domestic and international firms to identify opportunities to more closely integrate the UAE into global supply chains for AI technology, facilitating export opportunities and transfer of technology.

Automated logistics

The UAE government will work with private firms to develop and test pilot AI programs to manage logistics systems and optimize delivery chains.

Possible initiatives to realise four key opportunities to boost industry performance

B Boost industry performance

1. Support trial applications of AI systems to improve analytics and optimisation

- **Smart energy:** e.g. demand management, network maintenance and improved operation, including for emerging renewable energy systems
- **Smart financial services:** e.g. enhance access to credit through analysis of broad data in analysing loan applications etc.
- **Smart logistics:** e.g. use AI systems to manage complex logistics chains
- **Tourism**: e.g. develop AI digital assistants to provide tailored information to visitors
- **Aviation**: e.g. optimize flight paths and air-traffic management
- **Automated CRM**: e.g. use AI systems for customer relationship management

2. Deploy autonomous systems and remote sensing in private industries

- **Automated factories:** e.g. automated manufacturing assembly lines
- **Maintenance and repair:** e.g. use of sensors and drones to make maintenance and repair of physical infrastructure more efficient
- **Construction:** e.g. automated scanning drones and sensors to monitor safety issues and worksite integrity
- **Smart transport:** e.g. self-driving road fleets

3. Identify and expand AI export opportunities

- **Market creation support:** provide grants and other financial assistance for firms to establish connections in international markets
- **Size export opportunity:** Review domestic and international markets to identify opportunities for local firms and market niches
- **Incentives for exporters:** Encourage firms to develop an export strategy by providing grants and financial incentives for exporting firms
- **Global value chain:** Encourage global firms to partner with international AI technology firms to foster greater links into global value chains and enable technology transfer from international firms

4. Attract foreign investment in the AI sector

- **Foreign investment conditions:** Identify key obstacles to foreign investment, including legal limitations, that can be removed to attract priority investments in AI technology
- **Attract MNCs:** Establish a focussed initiative to market local areas to be the preferred locations for global firms to establish as their regional HQ
- **Centers of excellence:** Establish local geographic areas with special incentives to attract AI investments
- **Marketing campaign:** Advertise local capabilities to establish an international reputation for AI proficiency and attract foreign investment
- **Branding**: Create a distinct international brand through identifying the most competitive features of the local AI industry

For future consideration *Possible February announcement*

Potential flagship initiatives

Develop training programs

The UAE government will support the development of a new generation of AI engineers by funding the training of 1,000 students over the next three years.

Skills roadmap

The UAE government will undertake a comprehensive study of skills requirements for the AI economy, and the country's current workforce skill profile.

Curriculum review

The UAE government will lead a review of secondary school curricula to identify opportunities to increase teaching of "21st century skills" including interpersonal management and creativity.

We have identified possible initiatives to realise five key opportunities to ensure workforce readiness

C Ensure workforce readiness

1. Identify and manage future workforce needs

- **Identify technical requirements:** Identify the demand for and supply of technical skills and qualifications required to develop, refine and operate AI systems
- **Skills roadmap:** Develop a coherent understanding of changing skills, including expanded need for complementary skills
- **Identify priority industries:** Identify the industries and occupations which are likely to be most affected by Artificial Intelligence
- **Job-matching:** Provide platforms which match workers displaced by AI to new industries

2. Promote workforce based training

- **Diagnose:** Identify workforce training needs caused by AI disruption
- **Best practice:** Maintain database of best practice in training delivery
- **Demonstrate workplace training:** Deliver training to public sector workers
- **Develop programs:** Create public resources to be used by firms to plan and deliver training
- **Encourage training:** Provide financial assistance for training programs
- **Retraining:** Provide financial support for workers who have been displaced by AI

3. Develop domestic talent through tertiary education and vocational training

- **Strategic funding:** Review national funding of education to better align with national strategy
- **Encourage priority skills:** Encourage students to enroll in courses aligned with national strategy
- **Technical vocation:** Provide funding incentives to teach technical skills pertinent to AI
- **Complementary vocations:** Provide funding incentives to teach complementary "21st century" skills
- **Update courses:** Align course content with latest market practice
- **Tertiary education:** Support Universities to deliver high-priority courses

4. Attract skilled talent through migration

- **Identify local gap:** Identify areas where domestic skills need to be complemented through immigration
- **Tailored visas:** Expand skilled visa categories to prioritize AI skills
- **Visa applications:** Streamline visa application process for AI talent
- **Support firms to attract talent:** Extend financial support to support firms seeking to attract key talent
- **International brand**: Invest in developing global brand to attract AI talent in strategic sectors

5. Ensure schools provide future-ready education

- **Curriculum review:** Increase teaching of "21st Century" skills including interpersonal management and creativity
- **Incorporate AI skills:** Increase exposure to AI systems in school
- **Educate teachers:** Train teachers to incorporate AI within their curriculum
- **Monitor skills:** Measure student performance in AI skills against world
- **Set targets:** National targets for high-priority skill areas such as computer literacy
- **Equipment and technology:** Fund purchase of AI tech.

For future consideration (dashed box) *Possible February announcement* (solid box)

Potential flagship initiatives

AI R&D horizon scan

The UAE government will undertake a survey of local R&D capacity, in consultation with key research institutions, industry, and government agencies, to identify priority areas to bring AI capability to the UAE.

Public data

The UAE government will promote innovation in machine learning and AI programs by making public datasets available to train, develop and test algorithms.

Attract key thinkers

The UAE government will develop a programme to attract key AI thinkers to visit the UAE to participate in intensive cutting edge workshops

Possible initiatives to realise five key opportunities to lift R&D capability

D Lift R&D capacity

1. Coordinate research strategy

- **AI R&D horizon scan:** Draft a national R&D strategy identifying current knowledge gaps & strategically important areas of further research
- **Optimize policy:** Review current funding mechanisms and criteria to align these with outcomes of national strategy
- **Financial support:** Introduce new grants and funding mechanisms to promote high-priority areas for AI R&D
- **Awards:** Establish national awards system to publicize and promote leading researchers

2. Build research capability and talent

- **Talent strategy:** National talent identification strategy to inform selection process for key researchers/ teams
- **Improve best practice:** Share best practice in identifying and attracting talent
- **Marketing campaign:** Increase awareness of AI tech. opportunities
- **Attract key thinkers:** Short-term stays to develop AI solutions
- **Support talent acquisition:** Help attract researchers through financial incentives or institutional support
- **Develop leading research teams:** Through scholarships and funding

3. Create common R&D infrastructure

- **Map research infrastructure:** Identify gaps to focus long-term investment
- **Public data:** Create and make available public datasets
- **Central data access:** Collate government data to enable cross-agency use
- **Library of research:** Create open-access library of AI research outcomes from public research
- **Cloud servers:** Local servers capable of being used for AI computing
- **Incentives to create infrastructure:** Financial incentives to create public infrastructure including datasets

4. Provide direct and indirect funding for R&D

- **Identify funding gaps:** Review existing funding mechanisms
- **Set research targets:** Government-wide target for AI R&D grants and funding
- **Tax credits:** Offset R&D expenditure
- **Matched funding program:** Co-invest into AI research with private funding
- **Loans to small organizations:** Provide concessional loans to small/ growing AI research organizations
- **Attract global research:** Provide tax incentives and reduce other obstacles for global firms to investing in local AI research

5. Strengthen commercialisation of AI technology

- **Mentoring:** Provide mentoring services for AI firms taking innovations to market
- **Incubator fund:** Provide capital to enable development of new AI applications by domestic firms
- **Public resources:** Create publicly accessible sources of information on commercialization process
- **Strategic public procurement:** Work with promising AI researchers and emerging technology to provide "first customer" support

For future consideration *Possible February announcement*

Potential flagship initiatives

Government cooperation

Create a standing interagency panel to identify emerging regulatory standards, ensure consistency in regulation, and share best practice.

Identify regulatory principles

The UAE government will proactively help develop best-practice regulatory principles to guide laws which govern the development and application of AI technology.

We have identified possible initiatives to realise five key opportunities to adopt effective governance and regulation

E Adopt effective governance and regulation

1. Regularly consult with business and society to monitor emerging AI trends

Government co-operation: Create a standing interagency panel to identify emerging regulatory standards, ensure consistency in regulation, and share best practice

Industry consultation: Create a standing industry consultation panel to identify priorities from the private sector

Civic consultation: Create a standing consultation forum to gather feedback from citizens and civic society

2. Design regulation that enables AI innovation

Identify regulatory principles: Identify general, best practice regulatory principles to guide AI laws

Increase government expertise: Enable sound regulatory decisions by upskilling Govt. agencies in AI issues

Test regulation: Select geographic areas or specific industries to test new regulatory approach

Update intellectual property laws: Review IP laws to ensure they are suitable for innovation in AI economy

Data standards: Create common data standards to facilitate exchange and cooperation

3. Create ethical standards for the application of AI

Attribution: Ensure actions and decisions of AI systems are clearly/ justly attributable

Discrimination laws: Guard against development of unfair bias by AI systems

Data collection and sharing: Articulate legal standards to govern data collection and privacy issues

Limiting dangerous applications: Identify AI applications which need to be constrained for ethical or social reasons

4. Engage in international cooperation to help set AI standards

International conventions: Participate in global efforts to set principles for regulation/ governance of AI technology

Cross-border AI: Work with global partners to monitor cross-border applications of AI

Industry standards: Provide input into international processes to establish technological standards for operating AI systems

AI systems testing: Work with close allies to test and refine national AI systems

5. Ensure appropriate cybersecurity capabilities

National cybersecurity strategy: Develop national strategy that brings together stakeholders to identify and addresses key risks

Standards for critical industries: Work with critical industries to develop cybersecurity standards

Support cybersecurity capacity: Provide financial assistance to support development and attraction of firms with key cybersecurity technology

For future consideration *Possible February announcement*